6 DGLAP

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\left\{\mathcal{P}_{gg}(\xi)\,\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\,\frac{d^3N_{\xi\,\rm LO}}{d^3k} - 2\,N_c\,\frac{(1-\xi)}{\xi}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\frac{d^3N^y_{\xi\,\rm LO}}{d^3k}\right\} \tag{6.1}$$

7 Running coupling

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{g^2\,\beta_0}{16\pi^2}\,\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\,\frac{d^3N_{\rm LO}}{d^3k} \tag{7.1}$$

8 Finite part

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\,\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&+\,\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]\times\\
&\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\,\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\end{aligned} \tag{8.1}$$

where:

$$\frac{d^3N_{\xi\,\rm LO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\,\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{8.2}$$

$$\frac{d^3N^y_{\xi\,\rm LO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y)}{(x-y)^2\,(x'-y)^2}\,\times\left[U^{e'c}(y)-U^{e'c}(x')\right]\times\,\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{8.3}$$

$$\frac{d^3N_{\rm LO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\,\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{8.4}$$